

Roll No.

END SEMESTER EXAM JUNE- 2025

Name of the Program : B.Tech.(E&C)

Semester : VIII

Name of the Course : Neural Network and Machine Learning

Course Code : TEC 856

Time : 3 hours

Maximum Marks : 100 Marks

Note :

- I. All questions are compulsory.
- II. Answer any two sub questions among a, b & c in each main question.
- III. Total marks in each main question are twenty
- IV. Each Question carries 10 Marks.

Q1	(20 Marks)	CO1
(a)	Explain the concept of Artificial Neural Network and its use for pattern recognition	
(b)	Explain the different types of learning methodologies for Neural Network	
(c)	Explain with examples use of Linear Regression	
Q2	(20 Marks)	CO2
(a)	With the pattern $S=[1,-1,-1,1]$, demonstrate the recognition capabilities of auto associative neural network	
(b)	With the pattern:target pair $s:t$, where $s=[1,-1,1,-1]:t=[-1,1,-1]$ demonstrate the working of hetero associative neural network	
(c)	Realise AND-NOT Operation using a Neural Network	
Q3	(20 Marks)	CO3
(a)	With example explain the use of Radial Basis Function Neural Network for classification	
(b)	With examples explain the use of Support Vector Machine for classification	
(c)	Explain any two activation functions	
Q4	(20 Marks)	CO4
(a)	Explain the MAXNET architecture and illustrate its use for pattern $X=[x_1,x_2,x_3,x_4]=[0.1,0.3,0.8,0.6]$	
(b)	Explain the use of Learning Vector Quantization Neural Network for four bit inputs and two clusters $S_1=[1,0,0,0]$ belongs to cluster C_1 $S_2=[0,1,0,0]$ belongs to cluster C_1 $S_3=[1,0,0,0]$ belongs to cluster C_1 $S_4=[0,0,0,1]$ belongs to cluster C_2 $S_5=[0,0,1,1]$ belongs to cluster C_2	
(c)	Explain the clustering method as applied to classification	
Q5	(20 Marks)	CO5
(a)	Explain Hopfield Neural Network with a suitable example	
(b)	Explain the Boltzmann machine architecture	
(c)	Explain the working of MAXNET	

